

(Q1)

$(2, 1), (3, 0)$

$c_1(2, 1) + c_2(3, 0) = (a, b)$

$2c_1 + 3c_2 = a$

$c_1 = b$

$\begin{bmatrix} 2 & 3 \\ 1 & 0 \end{bmatrix}$

"So basis."

"spans."

$\begin{bmatrix} 2 & 3 \\ 1 & 0 \end{bmatrix} \Rightarrow (-3) \text{ linear independent.}$

$(Q3) c_1(x^2+1) + c_2(x^2-1) + c_3(2x-1) = (a, b, c).$

$\rightarrow c_1 + c_2 + 0c_3 = a$

$\rightarrow 2c_3 = b \rightarrow c_3 = b/2$

$\rightarrow c_1 + c_2 - c_3 = c$

$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 2 \\ 1 & -1 & -1 \end{bmatrix}$

$1(2) - 1(-2)$

$4 \neq 0 \text{ so}$

"Basis"

$(Q2) c_1(3, 2, -4) + c_2(2, 5, b) + c_3(1, 4, 8) = (a, b, c)$

$3c_1 + 2c_2 + c_3 = a$

$c_1 + 5c_2 + 4c_3 = b$

$-4c_1 + 6c_2 + 8c_3 = c$

$\begin{bmatrix} 3 & 2 & 1 \\ 1 & 5 & 4 \\ -4 & 6 & 8 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix}$

$= 3(26) - 2(24) + 1(26)$

$48 - 48 + 26 = 26 \neq 0$

"So linearly indep."

"So basis."

$c_3 = a - 2c_2 - 3c_1$

$c_1 + 5c_2 + 4(a - 2c_2 - 3c_1) = b$

$a + 5c_2 + 4a - 8c_2 - 12c_1 = b$

$-11c_1 - 3c_2 = b - 4a$

$11c_1 + 3c_2 = 4a - b \rightarrow (1)$

$-4c_1 + 6c_2 + 8(a - 2c_2 - 3c_1) = c$

$-4c_1 + 6c_2 + 8a - 16c_2 - 24c_1 = c$

$-28c_1 - 10c_2 = c - 8a$

$28c_1 + 10c_2 = 8a - c \rightarrow (2)$

"So span"

$= -1 \begin{bmatrix} 3 & 0 & 1 \\ 3 & -12 & -3 \\ -6 & -4 & 2 \end{bmatrix} + 1 \begin{bmatrix} 3 & 0 & 1 \\ 6 & -8 & 0 \\ -6 & -4 & 2 \end{bmatrix}$

$\begin{bmatrix} 3 & 0 & 0 & 1 \\ 6 & -1 & -8 & 0 \\ 3 & -1 & -12 & -1 \\ -6 & 0 & -4 & 2 \end{bmatrix}$

$(Q3) c_1(1+x) + c_2(1-x) + c_3(1-x^2) = (a, b, c)$

$c_1 + c_2 + c_3 = 0 \rightarrow$

$xc_1 + xc_2 = 0$

$0 + 0 - c_3x^2 = 0$

$c_3 = 0$

"So basis"

"So basis"

$\begin{bmatrix} 3 & 0 & 0 & 1 \\ 6 & -1 & -8 & 0 \\ 3 & -1 & -12 & -1 \\ -6 & 0 & -4 & 2 \end{bmatrix}$

$-4c_3 + b - a = c$

$4c_3 = b - a - c$

$c_3 = \frac{b-a-c}{4} \text{ so } c_1, c_2, c_4 \text{ so span}$

$(Q4) \begin{bmatrix} 3 & 6 \\ 3 & -6 \end{bmatrix} + \begin{bmatrix} 0 & -1 \\ 4 & -1 & 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 & -8 \\ -12 & -4 \end{bmatrix} + c_4 \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$3c_1 + 4c_4 = a \rightarrow c_4 = \frac{a-3c_1}{4}$

$6c_1 - c_2 - 8c_3 = b \rightarrow c_2 = 6c_1 - 8c_3 - b$

$3c_1 - c_2 - 12c_3 - c_4 = c$

$-6c_1 - 4c_3 + 2c_4 = d$

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Q7). $c_1(2, -3, 1) + c_2(4, 1, 1) + c_3(0, -7, 7) = (a, b, c)$

1a). $2c_1 + 4c_2 = a \rightarrow (1)$

$-3c_1 + c_2 - 7c_3 = b \rightarrow -3c_1 + c_2 - 7(c - c_1 - c_2) = b$

$c_1 + c_2 + c_3 = c \rightarrow -3c_1 + c_2 - 7c + 7c_1 + 7c_2 = b$

$c_3 = c - c_1 - c_2$

$4c_1 + 8c_2 = b + 7c$

$2c_1 + 4c_2 = a$

$2c_1 = \frac{a - 4c_2}{2}$

$4\left(\frac{a - 4c_2}{2}\right) + 8c_2 = b + 7c$

$0 \neq 0$ so can't span so not basis

1b). $c_1(1, 6, 4) + c_2(2, 4, -1) + c_3(-1, 2, 5) = (a, b, c)$

$c_1 + 2c_2 - c_3 = a \rightarrow c_1 = a + c_3 - 2c_2$

$6c_1 + 4c_2 + 2c_3 = b \rightarrow 6(a + c_3 - 2c_2) + 4c_2 + 2c_3 = b$

$4c_1 - c_2 + 5c_3 = c \rightarrow 4(a + c_3 - 2c_2) - c_2 + 5c_3 = c$

$6a + 6c_3 - 12c_2 + 4c_2 + 2c_3 = b$

$8c_3 - 8c_2 = b - 6a \rightarrow (2)$

$4a + 4c_3 - 8c_2 - c_2 + 5c_3 = c$

$6\left(\frac{c - 4a + 9c_2}{9}\right) - 8c_2 = b - 6a \rightarrow 9c_3 - 9c_2 = c - 4a \rightarrow (2)$

$c_3 = \frac{c - 4a + 9c_2}{9}$

$8c - 32a + 72c_2 - 8c_2 =$

$0 \neq 0$ so can't span. so not basis

Q8). $c_1(1 - 3x + 2x^2) + c_2(1 + x + 4x^2) + c_3(1 - 7x + 0x^2) = (a, b, c)$

$c_1 + c_2 + c_3 = a \rightarrow \left(\frac{c - 4c_2}{2}\right) + c_2 + c_3 = a$

$-3c_1 + c_2 - 7c_3 = b \rightarrow -3\left(\frac{c - 4c_2}{2}\right) + c_2 - 7c_3 = b$

$2c_1 + 4c_2 = c$

$= \frac{c - 4c_2 + 2c_2 + 2c_3}{2} = a$

$c_1 + 2c_2 = \frac{c}{2} \rightarrow (1)$

$c_1 = \frac{c - 4c_2}{2}$

$c - 4c_2 + 2c_3 = 2a$

Q9) done

(b)?

Q10). $c_1(\cos^2 x) + c_2(\sin^2 x) + c_3(\cos 2x) = (a, b, c)$

(a). $c_1(\cos^2 x) + c_2(\sin^2 x) = a$

$c_3(\cos 2x) = b$ Not a basis

$0 = c$

Q11). (a). $c_1(2, -4) + c_2(3, 8) = (1, 1)$

$2c_1 - 4c_2 = 1 \rightarrow c_1 = \frac{1 + 4c_2}{2}$

$-4c_1 + 8c_2 = 1$

$-4\left(\frac{1 + 4c_2}{2}\right) + 8c_2 = 1$

$W_2 = \left(\frac{5/28}{3/14}\right) \rightarrow \text{coordinate.}$

$\left(\frac{5}{28}, \frac{3}{14}\right) \rightarrow \text{coordinate.}$

$14c_2 = 3 \rightarrow c_2 = 3/14$

Q11) (b). $c_1(1, 1) + c_2(0, 2) = (a, b)$

$c_1 = a$

$\{W\}_3 = \begin{bmatrix} a \\ b - \frac{c}{2} \end{bmatrix}$

$c_1 + 2c_2 = b$

$a + 2c_2 = b$ so span

$c_2 = \frac{b - a}{2}$

Q12) done. (a). $(2, -1, 3) = c_1(1, 0, 0) + c_2(2, 2, 0) + c_3(3, 3, 3)$

$2 = c_1 + 2c_2 + 3c_3 \rightarrow c_1 = 2 - 2c_2 - 3c_3$

$-1 = 2c_2 + 3c_3 \rightarrow c_2 = -2$

$3 = 3c_3 \rightarrow c_3 = 1$

$c_3 = 1$

$2c_3 - 2\left(\frac{2b + 3c + 14c_3}{14}\right) = 2a - c$

c_3 will cancel so can't span

" so not basis "

$14c_2 - 14c_3 = 2b + 3c$

GENIUS $c_2 = \frac{2b + 3c + 14c_3}{14}$

Q14). $(4 - 3x + x^2) = p_1(1, 0, 0) + p_2(0, 1, 0) + p_3(0, 0, 1)$

(a) $4 = p_1, -3 = p_2, 1 = p_3$
 $\Rightarrow \begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix}$ Ans.

Q15) Done.

Q15) $c_1 \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} + c_3 \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} + c_4 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$

$c_1 = 1$ so $c_1 = 1, c_2 = -1, c_4 = -1, c_3 = 1$

$c_1 + c_2 = 0 \rightarrow c_1 = -c_2$ so "Span" Q16) Done

$c_1 + c_2 + c_3 = 1 \rightarrow c_3 = 2$ is "basis"

$c_1 + c_2 + c_3 + c_4 = 0$

$\begin{bmatrix} 1 \\ -1 \\ 2 \\ -1 \end{bmatrix}$ coordinate

Q17). $c_1(1+x+x^2) + c_2(0+x+x^2) + c_3(0+0+x^2) = (7-x+2x^2)$

$c_1 = 7$

$c_1 + c_2 = -1 \rightarrow c_2 = -8$ solve coordinate

$c_1 + c_2 + c_3 = 2 \rightarrow c_3 = 3$

$\begin{bmatrix} 7 \\ -8 \\ 3 \end{bmatrix}$

Q18) Done

Q19).

Q20). As Δ determinant 0 so not linear independent & not basis.

Q21). (a). $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -3 \\ -1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

$c_1(1, 0, -1) + c_2(1, 1, 2) + c_3(1, -3, 0) = (0, 0, 0)$

$c_1 + c_2 + c_3 = 0$

$c_2 - 3c_3 = 0$

$-c_1 + 2c_2 = 0$

$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -3 \\ -1 & 2 & 0 \end{bmatrix} = 1(1) - 1(-3) + 1(3)$

so independent

$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -3 \\ -1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$

Q21)(b) done

Q22)

$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -3 \\ -1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -3 \\ 0 \end{bmatrix}$

Q22)(a). $\begin{bmatrix} 2 & -1 & 0 \\ 1 & 1 & 1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \\ 0 \end{bmatrix}$

$4 = c_1 + c_3$

$c_2 = -c_3$

$(1, -2, 0) = c_1(1, 1, 0) + c_2(0, 1, 1) + c_3(1, 1, 1)$

$-2 = c_1 + c_2 + c_3$
 $0 = c_2 + c_3$
 $c_1 = -2, c_3 = 0, c_2 = -2$
 so coordinate $\rightarrow$

$\begin{bmatrix} -2 \\ -2 \\ 0 \end{bmatrix}$ GENIUS